

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2023/2024

JANUARY EXAMINATION

COMPUTER SCIENCE BACS2163
SOFTWARE ENGINEERING

TUESDAY, 9 JANUARY 2024

TIME: 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS) IN DATA SCIENCE
BACHELOR OF COMPUTER SCIENCE (HONOURS) IN INTERACTIVE SOFTWARE
TECHNOLOGY

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS
DEVELOPMENT

BACHELOR OF SOFTWARE ENGINEERING (HONOURS)

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

BACS2163 SOFTWARE ENGINEERING**Question 1**

- a) ABC Tech Solutions is a software development company specialising in enterprise-level software solutions. They were working on a large-scale project for a client in the healthcare industry. The client had complex requirements that were subject to change as the project progressed. To address these challenges and ensure early delivery of good functionalities, ABC Tech Solutions decided to adopt the incremental software development model.
- (i) Based on the above statement, illustrate and describe the **FOUR (4)** processes involved in incremental software development model. (8 + 12 marks)
- (ii) Outline **FIVE (5)** benefits that both the client and the software company can gain from the incremental software development model. (5 marks)

[Total: 25 marks]

Question 2

- a) You, as an appointed software engineer, are in-charge for designing an online management system for Focus Point Tutor center. Currently the tuition centre has been using manual way to manage the centre and now has decided to automate all their registration, classes scheduling, attendance, and payment processes. Design **FOUR (4)** sets of aperiodic stimuli-response for Focus Point Tutor online management system as mentioned. (8 marks)
- b) Managing a project group is one of the challenging and critical management tasks in software development projects. List and explain the **FOUR (4)** factors which influence the group working. (8 marks)
- c) Answer the following questions based on below statements shown in **Figure 1**.

Both functional and non-functional requirements can be formalised in the software requirements.

Source: <https://www.altexsoft.com/blog/functional-and-non-functional-requirements>

Figure 1: Statement

- (i) Define software requirements. (3 marks)
- (ii) Differentiate between functional and non-functional requirements in software requirement. (4 marks)
- (iii) Provide **ONE (1)** example for each type of requirement stated in **Question 2 c) (ii)**. (2 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 3**

- a) Answer the following questions based on given conversation shown in **Figure 2**.



Source: <https://languagelog.ldc.upenn.edu/>

Figure 2: Conversation

- (i) Identify the **FOUR (4)** design issues encountered by the developers when designing the user interface. (4 marks)
 - (ii) Identify **SIX (6)** essential user interface design principles and explain how they help resolve the issues faced by the developers. (12 marks)
- b) Based on the conversation shown in **Figure 3**, analyse the **THREE (3)** significant concern that the project manager should include in his project plan. (9 marks)



Source: <https://www.modernanalyst.com/>

Figure 3: Conversation

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 4**

- a) AC Web Solutions is a web development company specialising in e-commerce website development projects. They were tasked with creating an e-commerce platform for a fashion retailer. To ensure a holistic understanding of the system requirements, AC Web Solutions employed the viewpoint-oriented analysis approach.
- (i) Identify any **TWO (2)** advantages and **TWO (2)** disadvantages of using viewpoint-oriented analysis method in requirement gathering and analysis. (4 marks)
 - (ii) Identify any **SIX (6)** viewpoints of an e-commerce platform for a fashion retailer and prepare a viewpoint hierarchy. (12 marks)
- b) In a project management, a project manager would need to consider factors that could motivate employees throughout the project. Describe any **THREE (3)** levels of needs in Maslow's motivation model. (9 marks)

[Total: 25 marks]